

When should VEGO-AI involve a human?

A visual baseline from existing AirTravel evidence

WHAT WAS STUDIED

We can now see where an alert can be raised. We still cannot say whether the alert is correct.

Executive Summary

- VEGO-AI ON adds agent coordination, an inspectable Q&A log, and Detector-v1 labels.
- OFF is a direct workflow. It finished all cases with less time and cost in this run.
- Human labels are still missing. Quality, accuracy, benefit, and workload reduction are not measured.

Public external LLM-generated AirTravel corpus

12 seeded cases from 21 eligible

gpt-5.6-luna · PROVIDER_DEFAULT_NOT_SENT · one day

Not student data · not representative

3 inputs overlap Study 1 · no outputs reused or pooled

Provider-default stochastic run; a second run may differ.

Evidence ladder

ARCHIVAL-RETROSPECTIVE DESCRIPTIVE EVIDENCE

1 · Archival Study 1

4 purposive cases · 3 complete episodes · 3/3 candidate labels

ENGINEERING-ONLY FIXTURE

ENGINEERING_FIXTURE_NOT_SCIENTIFIC

2 · Engineering checks

9/9 rule modes · 724 fixture calls · 0 provider calls

PROSPECTIVE EMPIRICAL EVIDENCE

3 · Prospective comparison

12 matched AirTravel cases · ON versus OFF · ON STOPPED_AT_CAP

NOT_MEASURED

4 · Human validation

0 raters scored · alert correctness remains open

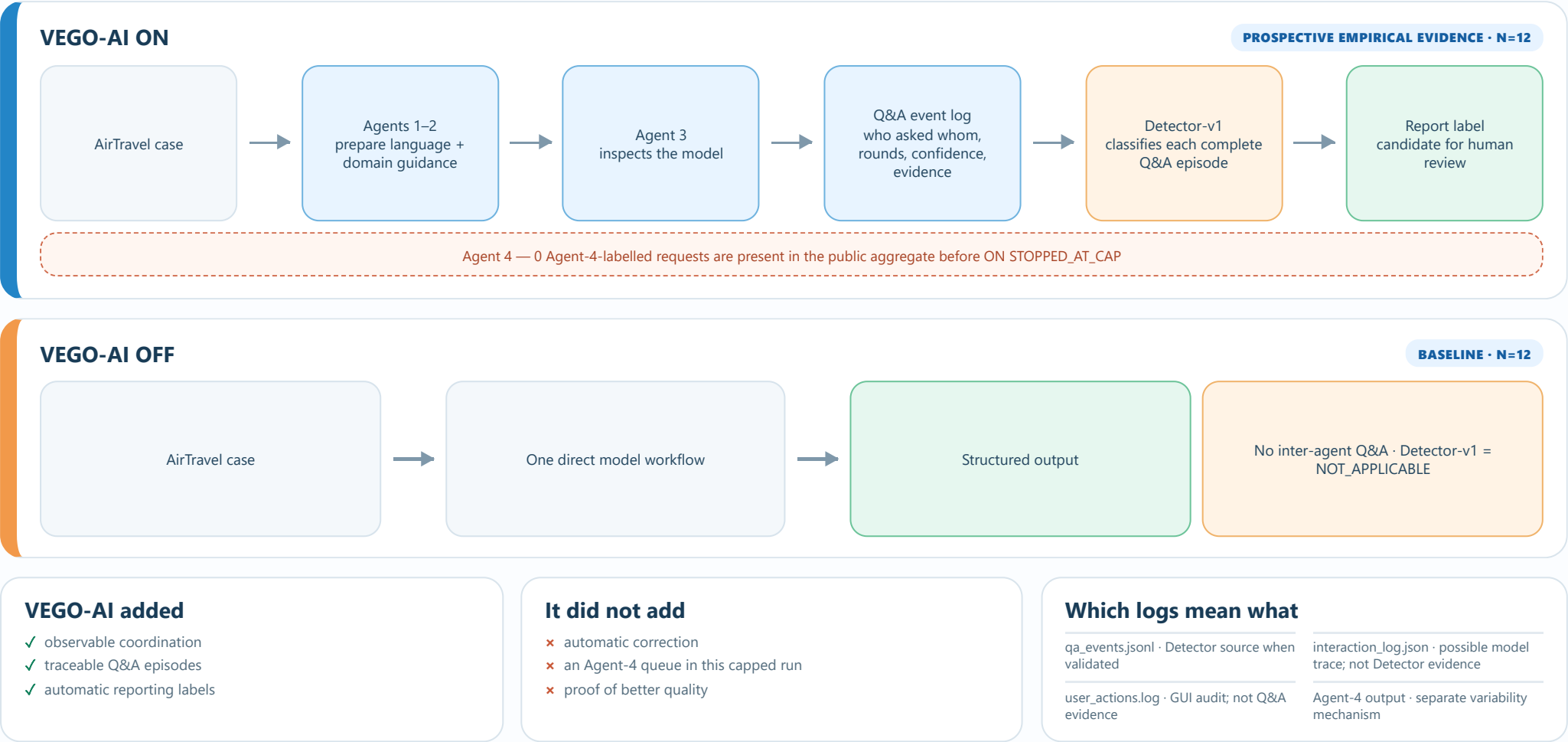
Run S2P-LIVE-20260908T192135Z

ON = STOPPED_AT_CAP

N = 12 planned cases

What changes when VEGO-AI is ON?

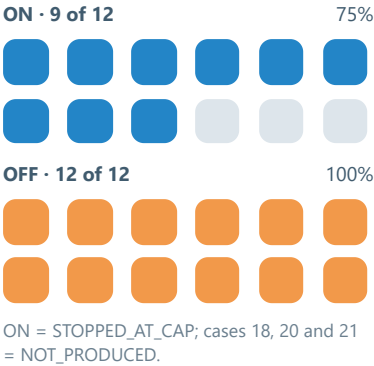
Same input, two different workflow packages



Operational baseline: ON versus OFF

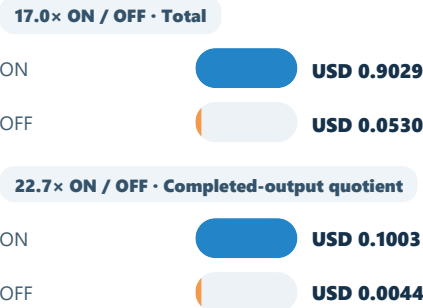
Descriptive measurements on 12 planned cases

Completed outputs



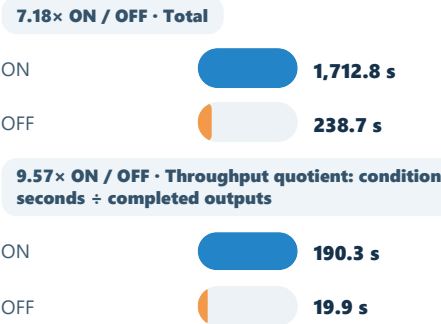
Source: analysis.json · **Condition:** ON + OFF
Denominator: 12 planned cases per condition
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: completed output artifacts ÷ planned cases
Limitation: ON stopped at cap; completion is not quality.

Recorded cost



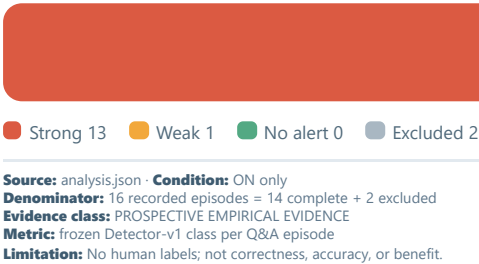
Source: analysis.json · **Condition:** ON + OFF
Denominator: all recorded requests; quotient uses ON 9 / OFF 12 completed outputs
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: recorded USD; quotient = condition cost ÷ completed outputs
Limitation: ON includes 2 shared setting-level requests and concurrency attributions; not isolated per-case cost or quality.

Elapsed time



Source: analysis.json · **Condition:** ON + OFF
Denominator: whole conditions; quotient uses ON 9 / OFF 12 completed outputs
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: condition wall-clock; throughput quotient = seconds ÷ completed outputs
Limitation: Not mean per-case latency; ON ran concurrently, with one model on one day.

Detector-v1 under ON



14/14 complete episodes received a candidate label. This is alert saturation—not proof of correctness.

Output structure check

ON 9/9 completed summaries were internally consistent; OFF 7/12. The workflows construct outputs differently.

NOT COMPARABLE AS QUALITY

Where can a human enter—and what is still missing?

Direct answer to Iris and Arnon’s questions

How the smart alert works

```
STRONG_ALERT = S1 OR S3 OR S7
WEAK_ALERT = no strong signal AND (S2 OR S6)
NO_ALERT otherwise
```

Alert = a candidate for human review in the report

Observed trigger signals · denominator: 14 complete ON episodes

S1 · at least one Low self-reported answer confidence	13/14	S2 · at least one Medium self-reported answer confidence	11/14
S3 · missing or zero-length evidence reference	0/14	S6 · more than one Q&A round	14/14
S7 · episode stopped at the maximum round	5/14		

Observed agent coverage

Asking agent	Answering agent	Q
Agent 3 · Model Inspector	Agent 1 · Language Advisor	64
Agent 3 · Model Inspector	Agent 2 · Domain Advisor	191

Agent 4 · Variability Explorer

Prospective aggregate: 0 Agent-4-labelled requests before ON STOPPED_AT_CAP

Separate archival path: EXECUTED_THEN_BLOCKED · queue

NOT_AVAILABLE

Source: analysis.json · Condition: ON only
Denominator: 255 questions; 250 answers
Evidence class: PROSPECTIVE EMPIRICAL EVIDENCE
Metric: question count by asking agent × answering agent
Limitation: Observed routes only; not agent performance.

Next human-validation gate

- 1 · Reconcile before scoring: current pack = 2 raters; supervisor freeze = 3 (Ali, Iris, Arnon).
- 2 · Adjudicate disagreements and record rationale + review time.
- 3 · Add valid non-alert examples before estimating recall or false negatives.

Human-labelled outcome

TP · alert + review needed	FP · alert + review not needed
NOT_MEASURED	NOT_MEASURED
FN · no alert + review needed	TN · no alert + review not needed
NOT_MEASURED	NOT_MEASURED

ENGINEERING-ONLY FIXTUREENGINEERING_FIXTURE_NOT_SCIENTIFIC · 9/9 rule modes · 724 fixture calls · 0 provider calls. Instrument behavior only.

DESCRIPTIVE INSTRUMENT ROBUSTNESS_ON_ONE_ACCEPTED_RUN500/500 seeded permutations + 3/3 named reorderings · one accepted run · 0 provider calls.

Current result: an inspectable escalation baseline—not validated human-intervention detection.